

6406533459031. ❌ All of these

MLF Quiz-2 PQ. Sept 24

Section Id : 64065373889
Section Number : 5
Section type : Online
Mandatory or Optional : Mandatory
Number of Questions : 12
Number of Questions to be attempted : 12
Section Marks : 40
Display Number Panel : Yes
Section Negative Marks : 0
Group All Questions : No
Enable Mark as Answered Mark for Review and Clear Response : No
Section Maximum Duration : 0
Section Minimum Duration : 0
Section Time In : Minutes
Maximum Instruction Time : 0
Sub-Section Number : 1
Sub-Section Id : 640653155064
Question Shuffling Allowed : No

Question Number : 64 Question Id : 6406531021289 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533459032. ✔ YES

6406533459033. ❌ NO

Sub-Section Number : 2
Sub-Section Id : 640653155065
Question Shuffling Allowed : Yes

Question Number : 65 Question Id : 6406531021290 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question *q = 5-2*

Consider the matrix

$$B = \begin{bmatrix} 4 & -2i & -1 \\ 2i & 0 & -i \\ -1 & i & 4 \end{bmatrix} \quad B^H = \begin{bmatrix} 4 & 2i & -1 \\ -2i & 0 & 0 \\ -1 & -i & 4 \end{bmatrix} \rightarrow B^H = \begin{bmatrix} 4 & -2i & -1 \\ 2i & 0 & -i \\ -1 & i & 4 \end{bmatrix}$$

Which of the following options is/are true with respect to B ?

Note: B^H is a conjugate transpose of B .

Options :

6406533459034. ✖ $B^2 = -B^H$

6406533459035. ✔ $B = B^H$ ✔

6406533459036. ✔ Every eigenvalue of B is real.

6406533459037. ✖ The determinant of B is zero.

$B = B^H$
then B is Hermitian.
∴ all e-values are real.

Question Number : 66 Question Id : 6406531021298 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a unitary matrix A . Let α be an eigenvalue of A , then which of the following options is/are true?

Options :

6406533459062. ✔ $|\alpha| = 1$.

6406533459063. ✔ α is an eigenvalue of A^H , where A^H is a hermitian matrix.

6406533459064. ✔ Eigenvectors of A are orthonormal.

6406533459065. ✖ A is not diagonalizable.

$$A \cdot A^* = A^* A = I.$$

e-value of unitary matrix have absolute value i.e 1.

∴ e-vectors corresponding to different e-values of unitary matrix are orthogonal.

Sub-Section Number :

Sub-Section Id :

Question Shuffling Allowed :

3

640653155066

Yes

Question Number : 67 Question Id : 6406531021291 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the following matrix:

$$A = \begin{bmatrix} 2 & -i & i \\ i & 2 & -i \\ -i & i & 2 \end{bmatrix}$$

$[11]$
 $Q \rightarrow$ e.vectors of $A^T A$ goes into Q .
 $Q^T \rightarrow$ e.vectors of AA^T goes into Q^T .

Diagonalize it as $A = Q\Lambda Q^T$, then Λ is given by

$\Lambda \rightarrow$ e-value

Options :

6406533459038. ✗

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3+\sqrt{2} & 0 \\ 0 & 0 & 3-\sqrt{2} \end{bmatrix}$$

Sum of e-value = sum of diagonal elem
 $\lambda_1 + \lambda_2 + \lambda_3 = 6$

6406533459039. ✓

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2+\sqrt{3} & 0 \\ 0 & 0 & 2-\sqrt{3} \end{bmatrix}$$

product of $\lambda_1, \lambda_2, \lambda_3 = \det |A|$.

6406533459040. ✗

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2+\sqrt{3} & 0 \\ 0 & 0 & 2-\sqrt{3} \end{bmatrix}$$

6406533459041. ✗ The given matrix is not diagonalizable.

Question Number : 68 Question Id : 6406531021296 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Suppose $A = \begin{bmatrix} 1+i & 2 \\ 0 & 3+i \end{bmatrix}$. Then, which of the following options represent the eigenvectors of A ?

Options :

6406533459054. ✓

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\lambda = 1+i$ $(A - \lambda I) \cdot v = 0$
 $(1+i) - (1+i) \quad 2$
 $0 \quad (3+i) - (1+i)$
 $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

6406533459055. ✗

$$\begin{bmatrix} 0 \\ -i \end{bmatrix}, \begin{bmatrix} i \\ i \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow y=0 \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \text{e vector corr. to } \lambda = 1+i$$

6406533459056. ✖

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$$\lambda = 3+i$$

$$\begin{bmatrix} (1+i)-(3+i) & 2 \\ 0 & (3+i)-(3+i) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow$$

6406533459057. ✖

$$\begin{bmatrix} -1 \\ -1 \end{bmatrix}, \begin{bmatrix} -i \\ -i \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow -2x + 2y = 0 \Rightarrow x = y \rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

e vector for $\lambda = 3+i$

Question Number : 69 Question Id : 6406531021297 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Given the quadratic form $Q(x, y, z) = 3x^2 + 2y^2 + z^2 + 4xy + 2xz + 4yz$.

expressed in matrix form as $Q(x, y, z) = \begin{bmatrix} x & y & z \end{bmatrix} M \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. Which of the following is M ?

Options :

6406533459058. ✔

$$\begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\nabla f(7, 4) = (18, -34)$$

6406533459059. ✖

$$\begin{bmatrix} 3 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

$$x_2 = x_1 - \frac{1}{2^2} (18, -34)$$

6406533459060. ✖

$$\begin{bmatrix} 2 & 3 & 1 \\ 3 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$x_2 = (7, 4) - (4.5, -8.5)$$

6406533459061. ✖

None of these.

$$x_2 = (2.5, 12.5)$$

Question Number : 70 Question Id : 6406531021299 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

$$\nabla f(2, 2) = [-6, -2]$$

$$\nabla f(x, y) = [6x - 6y, 2y - 6x]$$

Consider a function $f(x, y) = 3x^2 - 6xy + y^2 + 4$. Using gradient descent algorithm, with an initial guess of $(1, 2)$ and learning rate of $\frac{1}{1^2}$, where $t = 1, 2, 3, \dots$

What will be the value of (x, y) after two iterations?

Options :

6406533459066. ✖ $x = 7, y = 4$

6406533459067. ✔ $x = 2.5, y = 12.5$

6406533459068. ✖ $x = 4, y = 6$

6406533459069. ✖ $x = 3.5, y = 10$

$$x_{t+1} = x_t + \eta (-\nabla f(x_t))$$

$$x_1 = x_0 - \frac{1}{1^2} (-6, -2)$$

$$x_1 = (1, 2) - (-6, -2)$$

$$x_1 = (7, 4)$$

Question Number : 71 Question Id : 6406531021300 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Find the point on the line defined by $y = 2x + 1$ that is closest to the point $(4, 0)$.

Options :

6406533459070. ✖ $(3, 7)$

6406533459071. ✔ $(0.4, 1.8)$

6406533459072. ✖ $(0, 0)$

6406533459073. ✖ $(-1, 0)$

Sub-Section Number :

Sub-Section Id :

Question Shuffling Allowed :

$$m_1 \cdot m_2 = -1$$

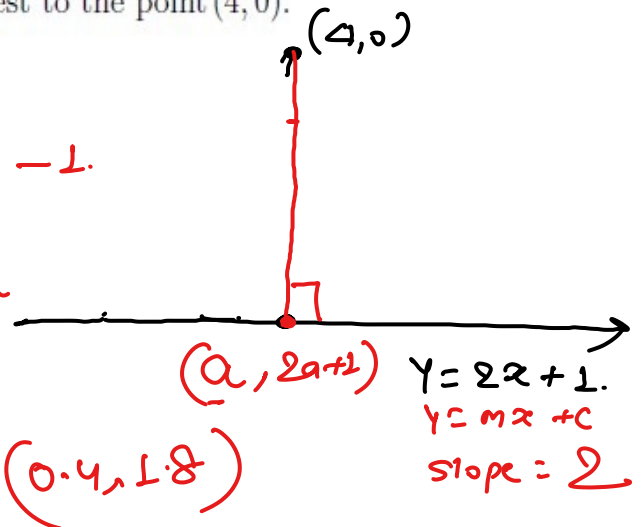
$$\frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{2a+1}{a-4} \times 2 = -1$$

$$= 4a + 2 = 4 - a$$

$$5a = 2$$

$$a = 0.4$$



Question Id : 6406531021292 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (72 to 74)

Question Label : Comprehension

Consider the matrix $A = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}$. Let the singular value decomposition

(SVD) form of A be written as $A = Q_1 \Sigma Q_2^T$.

$$2 \times 2 \rightarrow A A^T \rightarrow 3 \times 3$$

$$A \cdot A^T = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Based on the above data answer the given subquestions.

Sub questions

$$A A^T = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Question Number : 72 Question Id :

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following options is/are true?

Options :

6406533459042. ✖

$$Q_1 = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

6406533459043. ✔

$$Q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

6406533459044. ✖

$$Q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$$

6406533459045. ✖

$$Q_1 = \sqrt{2} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$$

Handwritten solution for Question 72:

$$A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \end{bmatrix}$$

$$(\lambda - 2)^2 - 1 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$(\lambda - 3)(\lambda - 1) = 0$$

$$\lambda = 1, 3$$

For $\lambda = 1$:

$$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x - y = 0$$

$$x = y$$

For $\lambda = 3$:

$$\begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x + y = 0$$

$$x = -y$$

Normalized eigenvectors:

For $\lambda = 1$: $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

For $\lambda = 3$: $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Question Number : 73 Question Id : 6406531021294 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following options is/are true?

Options :

6406533459046. ✖

$$Q_2^T = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 & -1 & 1 \\ 0 & \sqrt{3} & \sqrt{3} \\ -\sqrt{2} & 1 & \sqrt{2} \end{pmatrix}$$

6406533459047. ✖

$$Q_2^T = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 & -1 & 1 \\ 0 & \sqrt{3} & \sqrt{3} \\ -\sqrt{2} & -\sqrt{2} & -\sqrt{2} \end{pmatrix}$$

6406533459048. ✖

$$Q_2^T = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 & -1 & 1 \\ 0 & \sqrt{3} & \sqrt{3} \\ \sqrt{2} & -\sqrt{2} & \sqrt{2} \end{pmatrix}$$

6406533459049. ✔

Handwritten solution for Question 73:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

eigenvalues: 3, 1, 0

eigenvectors:

For $\lambda = 3$: $(2, -1, 1) \rightarrow \frac{1}{\sqrt{6}}(2, -1, 1)$

For $\lambda = 1$: $(0, 1, 1) \rightarrow \frac{1}{\sqrt{2}}(0, 1, 1) \rightarrow \frac{1}{\sqrt{6}}(0, \sqrt{3}, \sqrt{3})$

For $\lambda = 0$: $(-1, -1, 1) \rightarrow \frac{1}{\sqrt{3}}(-1, -1, 1) \rightarrow \frac{1}{\sqrt{6}}(-\sqrt{2}, -\sqrt{2}, \sqrt{2})$

$$Q_2^T = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 & -1 & 1 \\ 0 & \sqrt{3} & \sqrt{3} \\ -\sqrt{2} & -\sqrt{2} & \sqrt{2} \end{pmatrix}$$

$$\frac{1}{\sqrt{6}} \begin{bmatrix} 2 & -1 & 1 \\ 0 & \sqrt{3} & \sqrt{3} \\ -\sqrt{2} & \sqrt{2} & \sqrt{2} \end{bmatrix},$$

Question Number : 74 Question Id : 6406531021295 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following options is true?

Options :

6406533459050. ✓ $\Sigma = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

6406533459051. ✗ $\Sigma = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

6406533459052. ✗ $\Sigma = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

6406533459053. ✗ $\Sigma = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & \sqrt{3} & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Sub-Section Number :

5

Sub-Section Id :

640653155068

Question Shuffling Allowed :

No

Question Id : 6406531021301 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (75 to 77)

Question Label : Comprehension

Consider the following dataset:

$$\begin{matrix} x_1 \rightarrow \{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 6 \\ 5 \end{pmatrix} \} \\ \bar{x}_2 \end{matrix} \quad \begin{matrix} \bar{x}_1 = 4 \\ \bar{x}_2 = 3 \end{matrix} \quad \rightarrow \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

Suppose we want to project the above dataset onto a 1 dimensional space using PCA.

$$(2-4)(1-3) + 0 + 2 \times 2 \quad \text{Cov} = \frac{1}{3} \begin{bmatrix} 4+0+4 & 8 \\ 8 & 4+0+4 \end{bmatrix}$$

Based on the above data answer the given subquestions.

Sub questions

$$A \rightarrow 1$$

$$\frac{8}{3} \rightarrow \frac{8}{3} \times 2$$

$$\text{cov} = \frac{1}{3} \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}$$

Question Number : 75 Question Id : 6406531021302 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

The sample covariance matrix C for the given dataset is

Options :

$$C = \frac{1}{3} \begin{pmatrix} 8 & 8 \\ 8 & 8 \end{pmatrix}$$

6406533459074. ✓

$$C = \frac{1}{3} \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$$

6406533459075. ✗

$$C = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

6406533459076. ✗

$$C = \frac{1}{3} \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$$

6406533459077. ✗

$$\text{evalue: } 0, 16/3$$

$$\boxed{\lambda=2} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\boxed{\lambda=0} \text{ e-vector } \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$x=y \quad \text{evalue: } \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\tilde{x} = \left(\begin{bmatrix} 2 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (x \vee y) v_2$$

Question Number : 76 Question Id : 6406531021303 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

The eigenvalues of the covariance matrix are

Options :

$$\lambda_1 = 16, \lambda_2 = 0$$

6406533459078. ✗

$$\lambda_1 = 4, \lambda_2 = 0$$

6406533459079. ✗

$$\lambda_1 = 1, \lambda_2 = 0$$

6406533459080. ✗

$$\lambda_1 = \frac{16}{3}, \lambda_2 = 0$$

6406533459081. ✓

$$\tilde{x}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\tilde{x}_2 = \left(\begin{bmatrix} 4 & 3 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \left(\begin{bmatrix} 4 & 3 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \frac{7}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 7/2 \\ 2/2 \end{bmatrix} + \begin{bmatrix} 1/2 \\ -1/2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

Question Number : 77 Question Id : 6406531021304 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

If $\tilde{x}_1, \tilde{x}_2$ and $\tilde{x}_3$ are the projections of the data points onto the first principal component, then which among the following is correct?

$\tilde{x}_3 =$

Options :

$$\tilde{x}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \tilde{x}_2 = \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \tilde{x}_3 = \begin{pmatrix} 6 \\ 5 \end{pmatrix}$$

6406533459082. ✓

$$\tilde{x}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \tilde{x}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \tilde{x}_3 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

6406533459083. ✗

$$\tilde{x}_1 = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}, \tilde{x}_2 = \begin{pmatrix} 2/2 \\ 2/2 \end{pmatrix}, \tilde{x}_3 = \begin{pmatrix} 3/2 \\ 1/2 \end{pmatrix}$$

6406533459084. ✗

$$\tilde{x}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \tilde{x}_2 = \begin{pmatrix} 4 \\ 0 \end{pmatrix}, \tilde{x}_3 = \begin{pmatrix} 6 \\ 0 \end{pmatrix}$$

6406533459085. ✗

Sub-Section Number :

6

Sub-Section Id :

640653155069

Question Shuffling Allowed :

Yes

Question Number : 78 Question Id : 6406531021305 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements is true about the given matrix M ?

$$M = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$$

$\det = 0$

$$\lambda_1 + \lambda_2 = 6$$

$$\lambda_1 \cdot \lambda_2 = 0$$

$$4, 0$$

Options :

6406533459086. ✗ M is an invertible matrix.

6406533459087. ✓ M is a singular matrix.

6406533459088. ✗ The determinant of M is 4.

6406533459089. ✓ M is diagonalizable.

Sub-Section Number :

7

Sub-Section Id :

640653155070

Question Shuffling Allowed :

Yes

Question Number : 79 Question Id : 6406531021306 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following statements is true about the matrix $A = \begin{pmatrix} 1 & 4 \\ 4 & 16 \end{pmatrix}$?

Options :

6406533459090. ✖ A is positive definite. ✓

6406533459091. ✓ A is positive semi-definite. ✓

6406533459092. ✖ A is negative definite.

6406533459093. ✖ A is negative semi-definite.

$a > 0, \det > 0$

$a > 0, \det = 0$

$a < 0, \det > 0$

$a < 0, \det = 0$

$\det = 16 - 16 = 0$

$\det < 0 \rightarrow \text{not def.}$

Java

Section Id :	64065373890
Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	16
Number of Questions to be attempted :	16
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653155071
Question Shuffling Allowed :	No

Question Number : 80 Question Id : 6406531021307 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : PROGRAMMING CONCEPTS USING JAVA (COMPUTER BASED EXAM)"